Using Analytics to Improve Patient Experience and Provider Staffing

Application to Cancer Care Facilities

Yashi Huang, Arik Senderovich, Yaron Shaposhnik





OVERVIEW

- Collaboration with the **Dana-Farber Cancer Institute** (DFCI)

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- Resulted in two working papers
  - Describe developed solutions
  - Study broader questions surrounding these problems



PATIENT-SIDE

Waiting time prediction



OPERATIONS-SIDE

Decision support tool for planning provider sessions

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Waiting time prediction

Focus of this talk



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Decision support tool for planning provider sessions

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- Resulted in two working papers
  - Describe developed solutions
  - Study broader questions surrounding these problems
- Agenda
  - Additional background
  - Research questions
  - Results
  - Takeaways



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Waiting time prediction

Focus of this talk



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DANA-FARBER CANCER INSTITUTE



556

physicians

1,063

registered nurses

384,094

outpatient visits in 2023

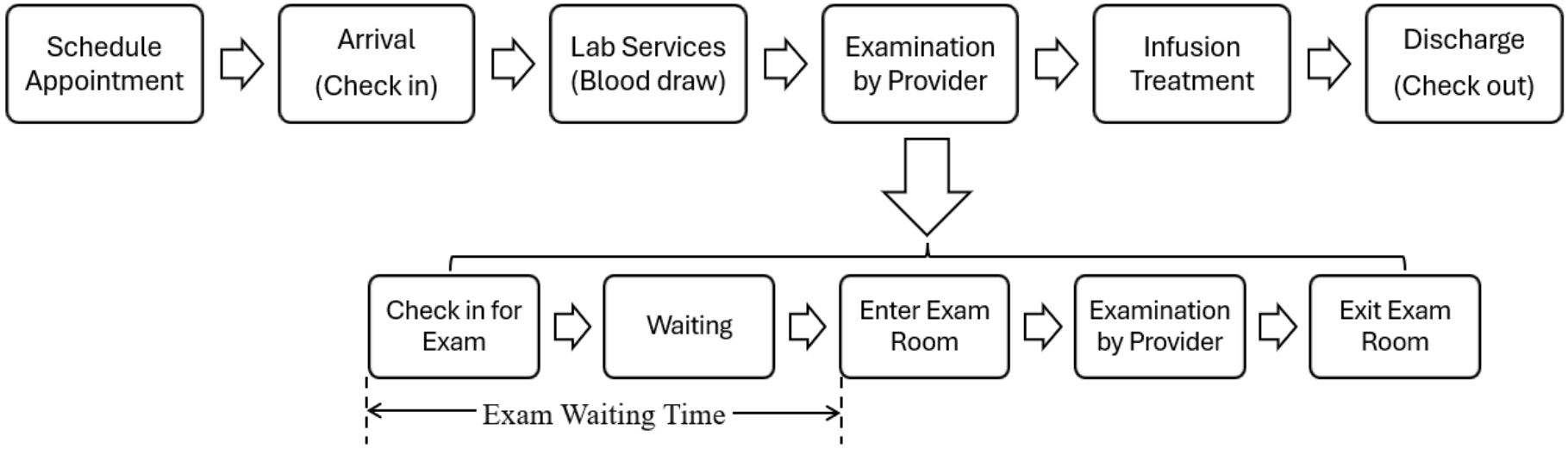
96,288

unique patients in 2023

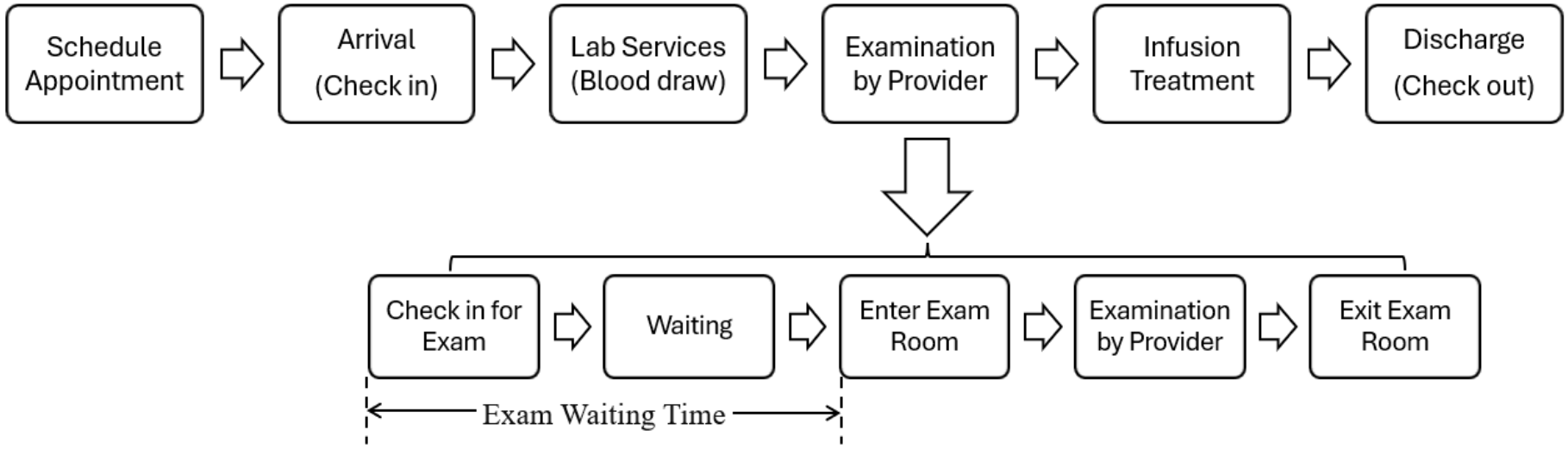
- One of 73 National Cancer Institute–designated cancer centers
- Adult and pediatric cancers and blood disorders
- Patients follow personalized treatment plans, scheduled in advance

Sources: Dana-Farber Facts 2024; NCI, Find an NCI-Designated Cancer Center

PATIENT FLOW AND DATA



PATIENT FLOW AND DATA



DATA RECORDED

Scheduling data (EHR)

Planned start times, durations, providers, and locations

Real-time locating (RTLS)

Badges track patients and staff: actual arrival, start, and end times

Data limitations

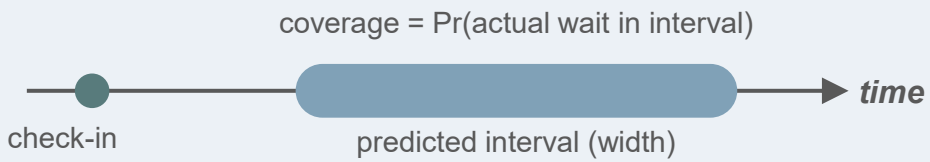
Noisy tracking (e.g., badges left behind); operational data only

RESEARCH QUESTIONS

THE PREDICTION TASK

Predict a patient's wait from check-in until entering the exam room

- **Interval**: lower and upper bound
- **Dynamic**: updated while the patient waits
- **Real-time**: scheduling + RTLS data
- **Requirements**: ~70% coverage, ~30-min width

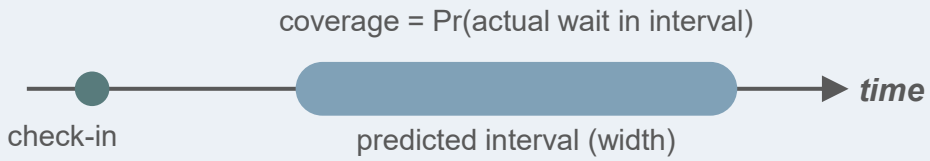


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RQ1 Which methods best balance the coverage-width tradeoff?

RQ2 What performance can be expected to be achieved in practice?

RQ3 What is the contribution of RTLS systems for waiting time prediction?

RQ4 Can the results be further improved?

BRIEF LITERATURE REVIEW

Foundations	Theory: Whitt (1999). Empirical: Batt & Terwiesch (2015), Dong et al. (2019). Behavioral: Liu et al. (2018). Survey: Ibrahim (2018)
Point prediction	Ibrahim & Whitt (2009), Senderovich et al. (2015), Ang et al. (2016), Kuo et al. (2020)
Probabilistic prediction	Ding et al. (2010), Salari et al. (2022), Arora et al. (2023) Fixed quantiles or discrete distributions
Interval prediction	Sun et al. (2012), Ataman & Sarıyer (2021), Nehme et al. (2022) Fixed quantiles, fixed category, simulation-based confidence interval
Appointment-based systems	Curtis et al. (2018), Lin et al. (2019), Al-Mousa et al. (2024): point predictions only

THIS WORK

- Dynamic, patient-specific intervals with width and coverage requirements
- Appointment-based setting with real-time (IoT) data
- First use of conformal prediction for waiting times
- Nested simulation to bound achievable accuracy
- Deployment and a series of pilots
- Applied contribution – study an important open problem that could impact millions of people

METHODS: FEATURES AND DATA

Data-driven approach: machine learning (ML) models trained on historical data



Feature engineering: predictors constructed from the scheduling and real-time data

Plan-based

- Slot order in provider's session
- Planned gap before appointment
- Planned durations, visit type

Real-time (RTLS)

- Arrival time vs. scheduled start
- Provider's in-service time
- Patients waiting ahead and their planned durations
- Upstream lab/infusion activity

Provider-specific

- Department
- Provider's tendency to start the first appointment late

Data split: chronological train / calibration / validation / test sets, Jan 2023 – Dec 2025 (encounters after preprocessing)



METHODS: MODELS AND DYNAMIC PREDICTIONS

Point models

- Linear regression
- Random forest
- XGBoost (XGB)

Quantile models

- Quantile regression
- ANN (pinball loss)
- RMTPP (recurrent)

Survival models

- Cox proportional hazards
- Conditional inference trees
- Random survival forest (RSF)
- NN-Survival

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Interval construction

- **Point models**: interval from the distribution of residuals
- **Quantile and survival models**: predicted lower and upper quantiles
- Quantiles chosen to minimize width subject to 70% coverage, per arrival group
- Calibrated on a held-out calibration set (conformal prediction)

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Dynamic predictions

- Re-predict every few minutes while the patient waits

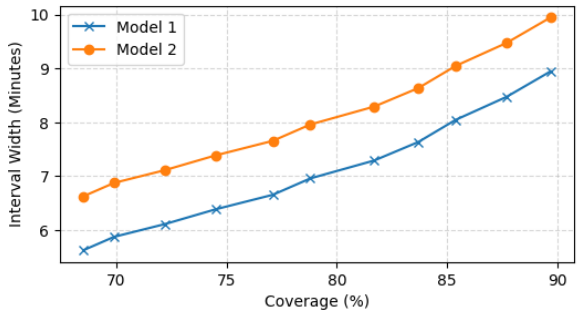
$$S(\theta \mid t_e; x_{te}) = \Pr(W \geq \theta \mid W \geq t_e, x_{te})$$

- Probabilistic models: updated features + time already waited
- Point models: elapsed time added as a feature

METHODS: EVALUATION METRICS

Coverage and width

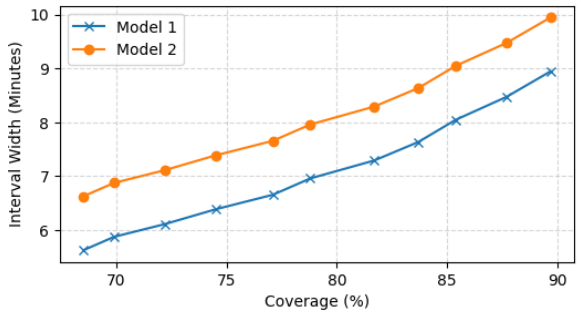
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- **Width**: upper − lower bound (target ≈30 min)
- Coverage–width curve compares models across targets



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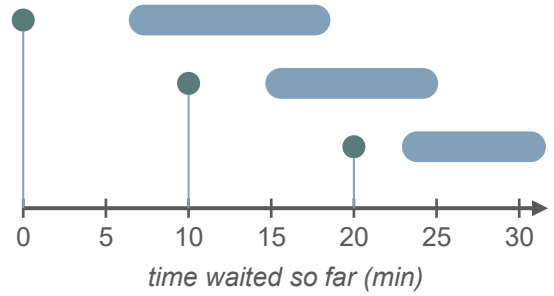
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Quality over time

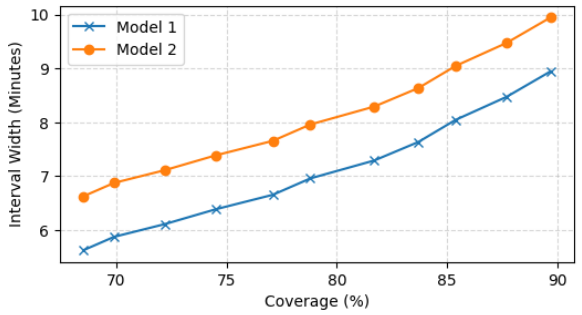
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- Only patients still waiting are included at each time
- Coverage and width of the remaining wait



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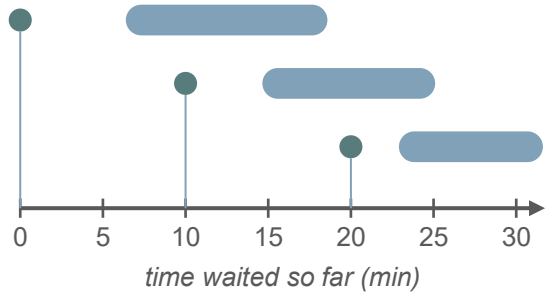
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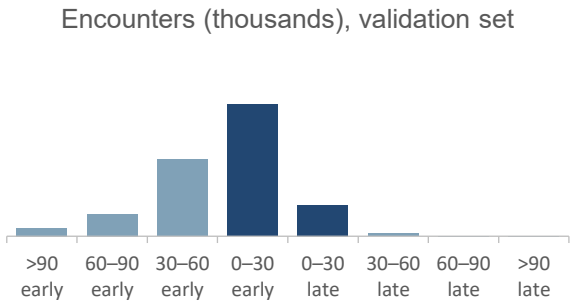
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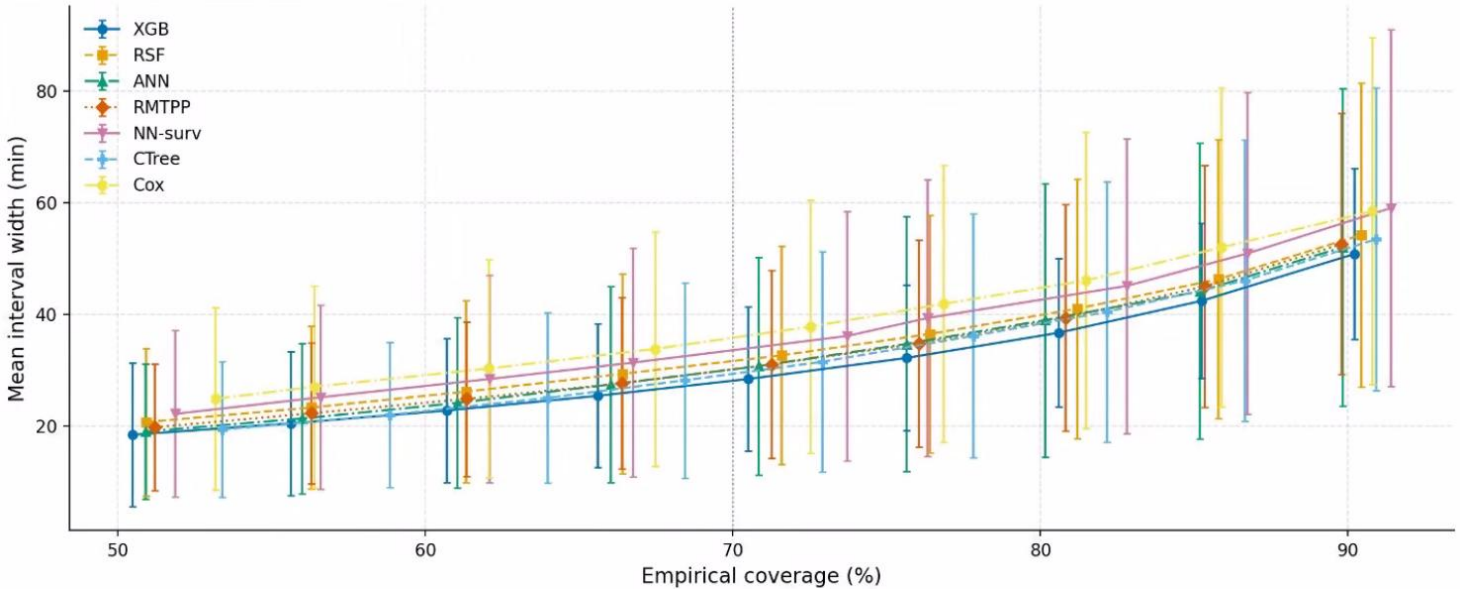
By arrival group

- Arrival time relative to the appointment (e.g., 0–30 min early)
- Intervals calibrated per group
- Within ±30 min of the appointment: primary use case (≈60% of patients)



RQ1: COVERAGE–WIDTH TRADEOFF

Which methods best balance the coverage-width tradeoff?



- **Cox and NN-Survival:** dominated
- **RSF, ANN, RMTPP, CTree:** nearly indistinguishable
- **XGB:** narrowest average width at every coverage level (≈2–4 min narrower at 70%)

28.4 min

mean width at 70% coverage (at arrival)

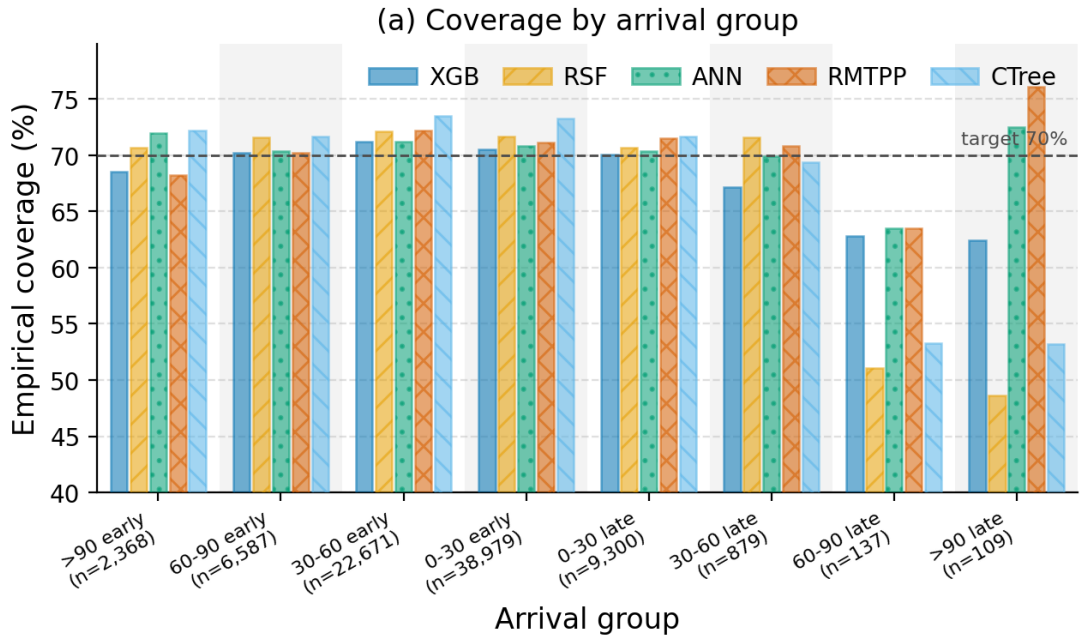
36.7 → 50.8 min

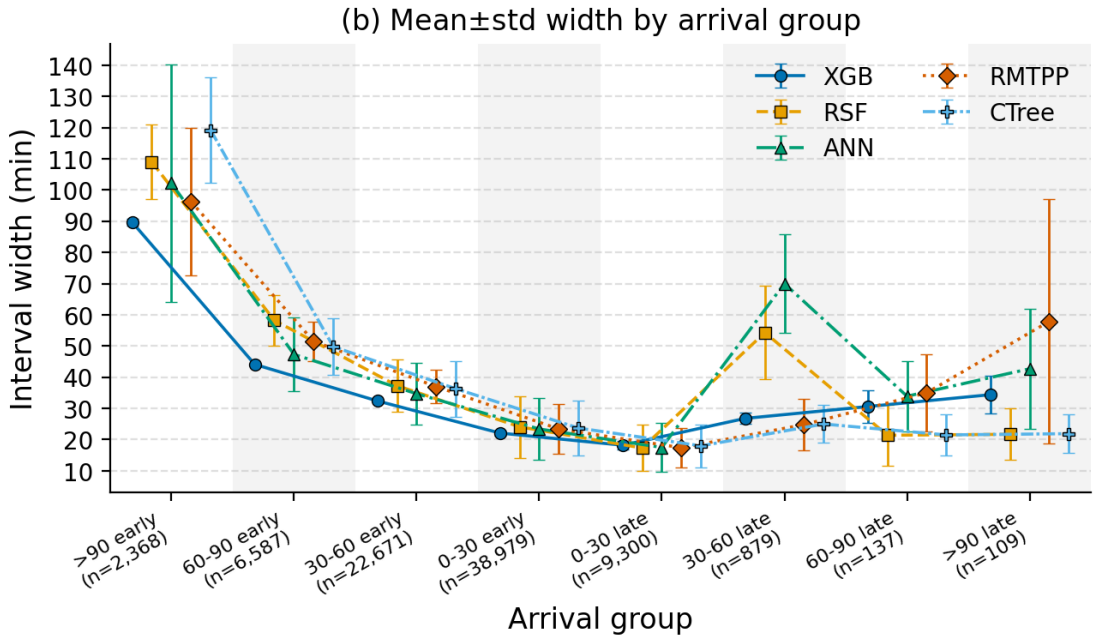
mean width at 80% → 90% coverage: higher coverage, wider intervals

Coverage–width curves at arrival, validation set; error bars show the spread of widths across patients (±1 sd)

RQ2: PERFORMANCE BY ARRIVAL GROUP

What performance can be expected to be achieved in practice?





≈88%

of patients can receive informative intervals at approximately 70% coverage

4.3%

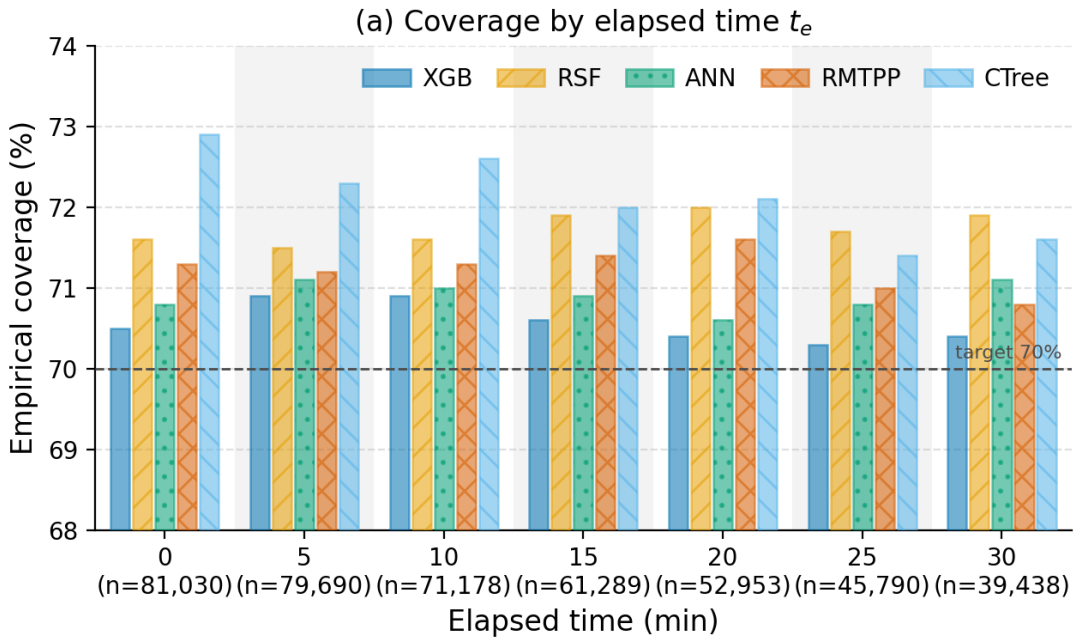
of patients arrive >90 min early or >30 min late, where intervals widen or coverage falls below target; one-sided intervals instead

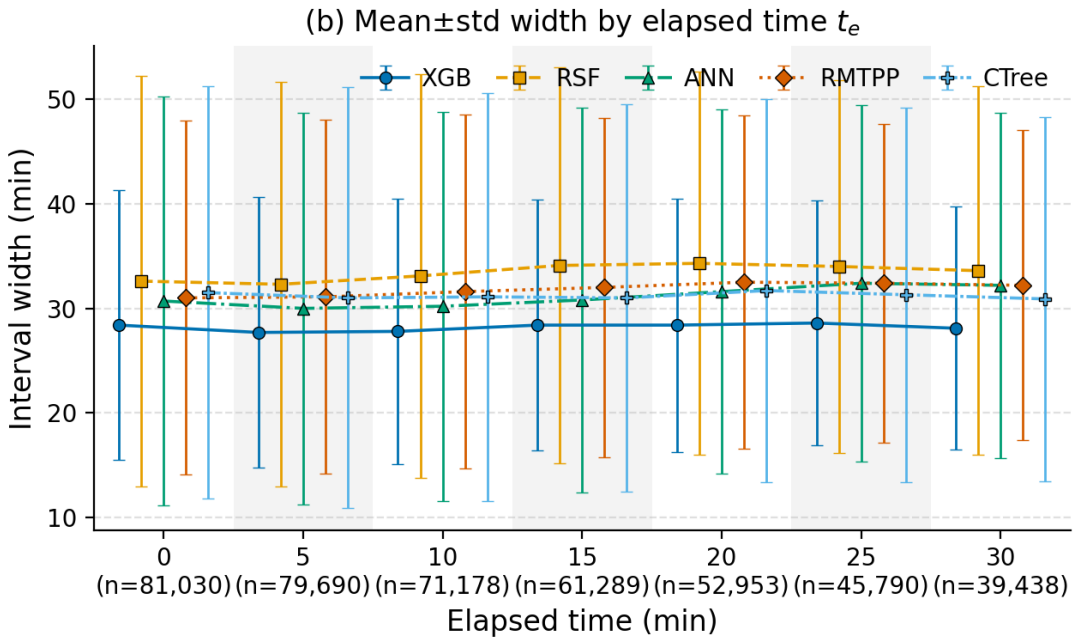
≤ 1.3 min

spread of XGB widths within an arrival group (others: 6–10 min): compact and stable, but barely personalized

RQ2: DYNAMIC PREDICTIONS

What performance can be expected to be achieved in practice?





70–73%

coverage for all models from arrival to 30 min of waiting

< 29 min

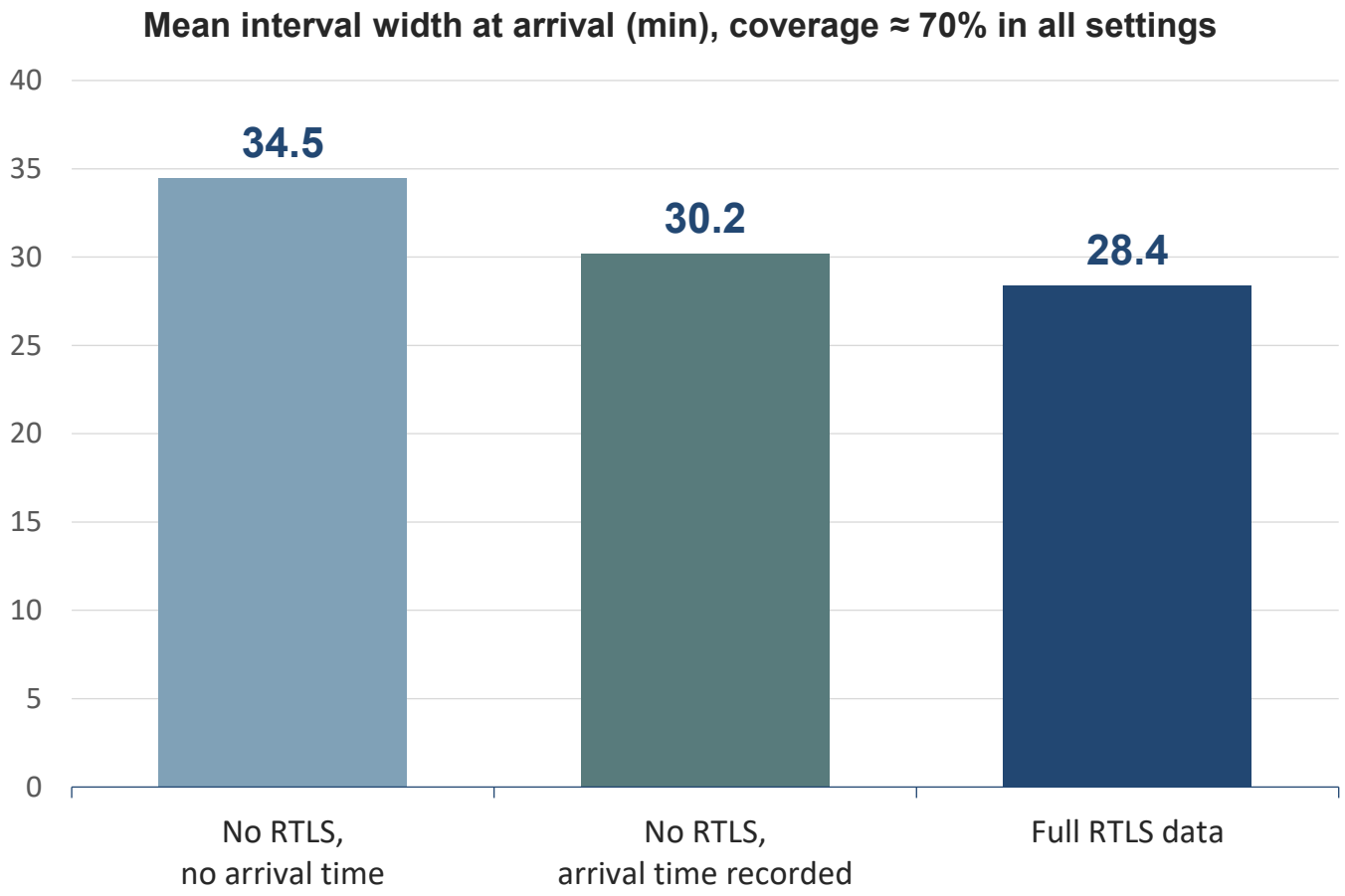
XGB widths throughout, 2–6 min narrower on average than the other models

Stable

widths don't shrink over time: patients still waiting are harder cases

RQ3: VALUE OF REAL-TIME DATA

What is the contribution of RTLS systems for waiting time prediction?



–6 min

narrower intervals with RTLS vs. schedule-only data

–4 min

of that gain comes from just recording arrival time

- RTLS data substantially improve predictions
- Hospitals without real-time tracking can still give useful predictions

RQ4: ESTIMATING THE IRREDUCIBLE ERROR

Can the results be further improved?

Are our intervals limited by the models, or by irreducible randomness in the service process?

Question: How can we estimate the irreducible error, i.e., the best accuracy any predictor could achieve?

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Answer: nested simulation

1. **Outer simulation:** simulates the clinic session (arrivals, services, queues)
2. **Pause at each trigger:** when an event calls for a prediction (e.g., a patient arrival)
3. **Inner simulations:** many independent continuations from the paused state estimate the waiting time
4. **Optimal prediction:** the best any predictor can do, i.e., an estimate of the irreducible error

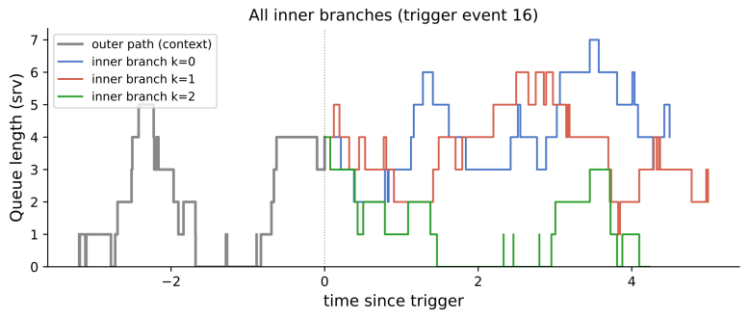


Illustration: inner simulations launched at one trigger event (M/M/1 queue)

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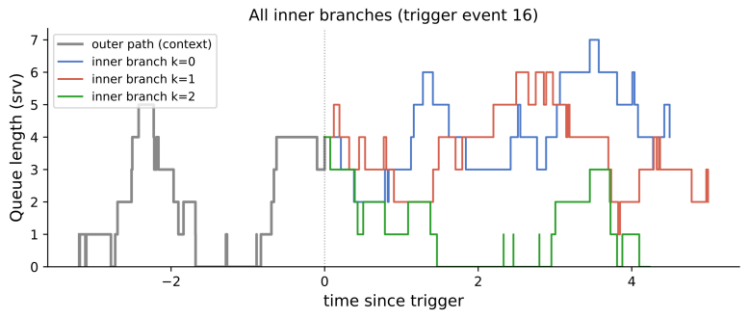
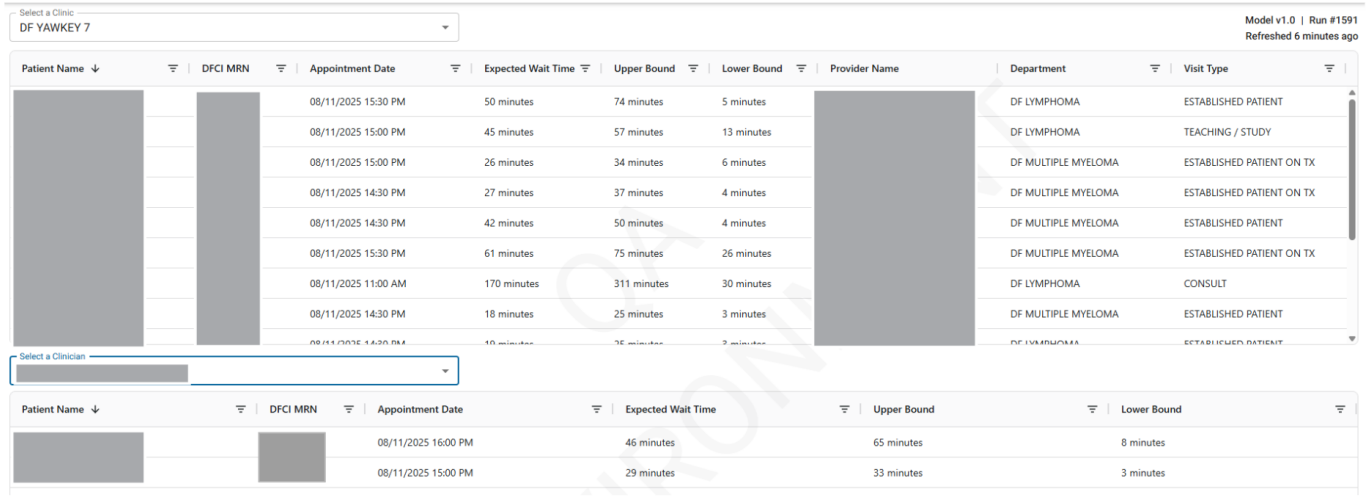


Illustration: inner simulations launched at one trigger event (M/M/1 queue)

Finding: in our simulation study, the irreducible error is substantial, on the order of the models' errors

DEPLOYMENT



Web application used by clinical assistants (identifiers hidden)

- First version based on RSF
- Real-time pipeline from EHR and RTLS data
- Model size cut from 20 GB to 2 GB
- Display refreshes every 5 minutes
- Data-drift and performance monitoring
- Half-intervals when too wide: “at least X min” (or “uncertain”)

Jun–Aug 2025

On-site development

Development and testing on site; by the end, used for “escalations”

Feb 27, 2026

Phase 1: “quiet” pilot

A month of staff-only access: is the app stable, used with every patient, easy to interpret, and useful?

Apr 17, 2026

Phase 2: sharing with patients

Clinic Assistants share each patient’s predicted wait after vitals are taken

Aug 31, 2026

Phase 3: waiting-room display

Predictions shown on a TV application in the waiting room

DEPLOYMENT: RESULTS

RQ2 in practice: *What performance can be expected to be achieved in practice?*

REAL-TIME PERFORMANCE (JUL–AUG 2025)

Elapsed (min)	Predictions	Coverage	Width (min)
0–5	1,563	79.9%	40.7
5–10	2,176	79.9%	40.5
10–15	1,930	79.5%	40.0
15–20	1,698	78.1%	39.3
20–25	1,499	77.9%	38.8
25–30	1,312	76.8%	37.4
30+	6,656	73.4%	35.2

PILOT FEEDBACK (PHASE 2)

91.0 vs. 87.6

patient-experience score on communication about delays (pilot, n = 108, vs. baseline, n = 3,397)

77%

of patients who recalled an estimate found it helpful (10/13)

69%

reported better communication about waits than at past visits (9/13)

NEXT STEPS

Of interest

DFCI has expressed interest

Broader use at DFCI

Other departments have expressed interest; the infrastructure can support new clinical settings

Beyond the exam step

Predict waits for blood work and infusion, and for the entire multi-stage visit

Possible

not currently pursued by DFCI

From information to decisions

Use predictions to prioritize patients dynamically and improve appointment scheduling

Richer data

Incorporate clinical data to reduce uncertainty about service durations

TAKEAWAYS

RQ1

Which methods best balance the coverage-width tradeoff?

XGB: narrowest intervals on average (28.4 min at 70% coverage), but not in every arrival group

RQ2

What performance can be expected to be achieved in practice?

Offline: ≈70% coverage with ≈28-min intervals. Deployed (RSF): 73–80% coverage with intervals of 35 to 41 min

RQ3

What is the contribution of RTLS systems for waiting time prediction?

RTLS narrows intervals by ≈6 min; recording arrival times alone captures ≈4 min of that gain

RQ4

Can the results be further improved?

Only marginally: an idealized optimal predictor does about as well, so much of the error is irreducible (unless more data is collected)

PRACTICAL TAKEAWAYS

Building the model

- **Use the schedule:** slot order, gaps, provider lateness
- **Tune the quantiles:** asymmetric bounds cut ≈5 min of width
- **Calibrate per arrival group:** a small lookup table keeps coverage on target
- **Simple can work:** XGB is narrowest on average, but check late arrivals

Communicating predictions

- **Intervals, not points:** much of the uncertainty is irreducible
- **Half-intervals when too wide:** “at least X min” or “uncertain”
- **Keep updating:** coverage holds while patients wait; widths don't shrink
- **Imperfect estimates still help:** 10 of 13 patients found them helpful

Deploying it

- **Start without tracking:** arrival times give ≈2/3 of the RTLS gain
- **Live ≠ back-test:** monitor coverage and width in production
- **Engineering matters:** model size, refresh, drift monitoring, versioning
- **XAI helped developers, not CAs:** it validated the model; CAs (arguably) lack the time and expertise to analyze predictions
- **Requirements evolve:** developing a single general model is beneficial
- **Surprisingly expensive:** staff hours and cloud computing

Decision-Support Tool for Staff Planning

Given a day's provider sessions, forecast chair demand throughout the day, at each infusion unit and across the center.

1 INPUT · PROVIDER SESSIONS

Example: one Monday

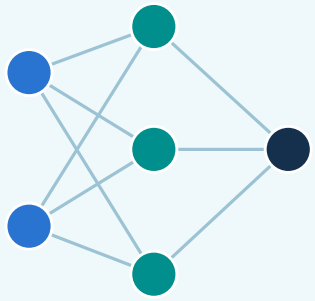
MONDAY

- Floor 1 — Dr. X — AM
- Floor 1 — Dr. Y — Full day
- Floor 2 — Dr. Z — PM

+ other scheduled provider sessions

↔ Input is a set, not an ordered sequence

2 PROBABILISTIC FORECASTING MODEL

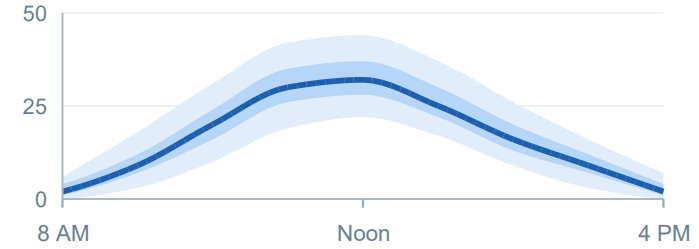


Sessions → Demand

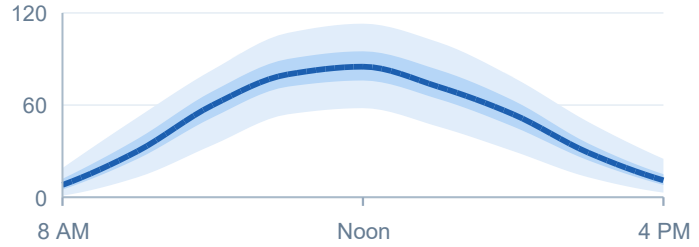
Predict expected demand and quantiles

3 OUTPUT · Demand CURVES

Floor 1· chair demand



Entire center · chair demand



Expected · 50% interval · 90% interval



THANK YOU

Questions?